

Youssef Elbehiry

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Summary

B.Sc. Mechanical Engineering graduate (UDE, Thesis Grade: 1.0, July 2026), enrolled in M.Sc. Mechanical Engineering at UDE from Winter Semester 2026/27. Hands-on experience in thermohydraulic systems and safety-relevant rotating equipment at a 500 MW combined-cycle power station. Strong foundation in thermodynamics, fluid dynamics and MATLAB/Simulink modelling. Skilled in mechanical design, 3D CAD, FEA simulation, GD&T and Lean process improvement.

Work Experience

Digital Trust & Safety Analyst — Telus Digital, Essen Part-time 20 h/week	11/2023 – Present
Evaluate 200+ digital content items per shift against safety policies; analyse risk patterns and proactively flag potential threats; maintain quality targets alongside full-time engineering studies and thesis.	
Industrial Internship — MEGA for Construction & Industries Cairo West Power Station	06–09/2025
500 MW combined-cycle plant, Egypt 13-week internship	
- Overhauled rotating equipment (boiler feed pumps, centrifugal pumps, heat exchangers) in a steam power plant; developed hands-on understanding of boiler, turbine, pump, heat exchanger, and GIS (gas-insulated switchgear) operation; ensured quality and safety compliance. - Applied vibration spectrum analysis for predictive maintenance diagnostics; conducted hydraulic performance testing to verify restoration to design specification; documented findings in structured technical reports.	

Education

B.Sc. Mechanical Engineering — University of Duisburg-Essen, Germany	10/2021 – 07/2026
GPA: 2.0 Thesis Grade: 1.0 Graduated: July 2026 M.Sc. Mechanical Engineering — enrolled from WS 2026/27	
Thesis: "Analysis of Influences on Roll Force, Torque and Forward Slip in Hot Flat Rolling of Steel by Fast Simulation" (03.07.2026)	
Modules: Thermodynamics, Fluid Dynamics, System Dynamics & Control (MATLAB/Simulink), Engineering Dynamics (FEM), Machine Design	
B.Sc. Mechanical Engineering — Mansoura University, Egypt	09/2017 – 07/2021
GPA: 1.3	
Modules: Manufacturing Processes, Machine Design, CAD, Engineering Drawing, Thermodynamics	

Skills

CAD & Design: SolidWorks – Advanced (CSWA Certified), AutoCAD – Intermediate, CATIA V5 – Intermediate, NX Siemens – Intermediate
Simulation & Analysis: MATLAB/Simulink – Intermediate, ANSYS – Intermediate, FEM – Intermediate
Measurement & Testing: Thermohydraulic performance testing; vibration spectrum analysis; dimensional inspection; GD&T
Lean & Quality: Lean Six Sigma White Belt
IT & Data: MS Office – Proficient, Python – Basic, C – Basic, SAP – Basic, Power BI – Basic
Soft Skills: Technical Documentation, Team Leadership, Structured Analytical Thinking, Cross-cultural Communication

Technical Projects

Bachelor Thesis (Grade: 1.0): Hot Flat Rolling Process Analysis & Simulation — UDE	2026
Evaluated six analytical rolling theories; validated against 52 measured passes on C55 steel across five parametric studies (pass reduction, roll-gap geometry, interstand tensions, friction, multistand rolling).	
HHO Electrolyzer Team Leader — Mansoura University	2020
Designed electrolyzer in SolidWorks; selected AISI 316L electrodes for corrosion resistance; conducted electrochemical efficiency measurements and validated hydrogen production against design targets. Led 4-person team to prototype.	
Dual-Axis Solar Tracking System Team Leader — Mansoura University	2019
Designed a dual-axis solar tracker using Arduino and 4 LDR sensors; modelled mechanical frame in SolidWorks. Led a team of 6 through design, servo actuation integration and prototype delivery.	

Certifications

CSWA – Certified SolidWorks Associate (Mechanical Design)	A Hands-on Introduction to Engineering Simulations
Lean Six Sigma White Belt	Introduction to Geometric Dimensioning and Tolerancing
Introduction to Composite Manufacturing	Simulink Onramp
MATLAB Onramp	German Language Certificate – B1

LANGUAGES

German: B2 • English: C1 • Arabic: Native